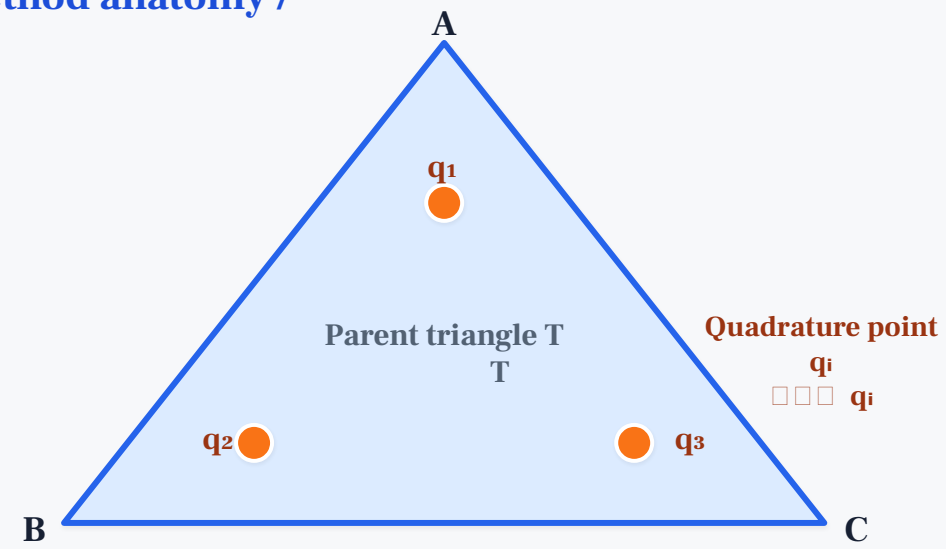
Three-point Gaussian quadrature for P–M analysis

P–M 3

The triangle stays unchanged. Three optimal points integrate every polynomial of degree ≤ 2 exactly.

. 2 .

Method anatomy /



Three-point degree-2 rule / 3 2

Barycentric coordinates / q_i = permutations of (⅔, ⅙, ⅙)

Integration weight / $w_i = A_t/3$

$$\int_t f\,dA \approx \Sigma_i\, w_i\, f(q_i)$$

P–M moment example / P–M

Normalized triangle / : A(0,0), B(1,0), C(0,1), $A_t = 1/2$

Linear stress / : $\sigma(x,y) = 1 + 2x$ Reference axis / : $x_0 = 0$

Each point / : $F_i = \sigma_i A_i$ and / $M_{Y,i} = F_i (x_i - x_0)$
With $A_i = A_t/3$: $P = \Sigma F_i = 5/6$ $M_Y = \Sigma F_i x_i = 1/3$

Method /	P	$M_Y = \Sigma(F_i \cdot d_i)$	M_Y error
1-point centroid 1	5/6	5/18 = 0.2778	-16.7%
3-point Gauss 3	5/6	1/3 = 0.3333	0%

Both methods recover P exactly. The moment exposes the missing second-order term.
P , 2 .

Why it matters / : The centroid rule gives M_Y 16.7% too low; 3-point Gauss is exact for this linear stress field.
1 M_Y 16.7% , 3 . Mesh refinement is still required across a neutral axis or material kink / . .